

README

Overview

The code in this replication package constructs the analysis file from source files provided by the West Midlands Police Force. One code file for STATA runs all of the code to generate the estimation data and to produce the 3 main figures, 6 main tables, 2 appendix figures, and 8 appendix tables in the paper. Moreover, auxiliary results mentioned in the paper are produced here as well. The replicator should expect the code to run for about 22 hours.

Data Availability and Provenance Statements

The data used in this article consist of administrative records provided by West Midlands Police (WMP) under a formal Information Sharing Agreement (ISA) between West Midlands Police, the University of Birmingham, and the ifo Institute, Munich (Version 3.0, May 2018).

The authors are secondary data users; the data were generated and held by West Midlands Police and accessed under the terms of the ISA. The data are stored at the ifo Institute, Munich (Poschingerstraße 5, 81679 München) on a secure, password-protected server with restricted access. Security measures include device-level encryption and continuous IT-breach monitoring by the ifo IT team. The data cannot be redistributed as part of a replication package. Access is restricted to parties named in the ISA, and personal data may not be published in identifiable form unless the individuals concerned have provided explicit consent.

Researchers wishing to replicate this work should contact the WMP Evidence Based Practice Team (our contact at the time of signing the ISA was Jenny Richards; evidencebasedpractice@west-midlands.pnn.police.uk; 0845 113 5000 Ext 801) to inquire about the conditions under which a new data-sharing agreement may be established. Any such access will be subject to WMP's data governance procedures, ethical approval, and the execution of a new ISA.

Auxiliary data used in this study (for two figures in the main text) comes from the British Crime Survey. To replicate the numbers, download the replication package (<https://onlinelibrary.wiley.com/doi/abs/10.1111/ecoj.12246>), obtain survey access, run the code in "BCS2005_EJ.do" and "Dataset_Final_EJ.do", and place "Analysis_Final_EJ.dta" in the Rawdata folder.

Statement about Rights

- ☒ I certify that the author(s) of the manuscript have legitimate access to and permission to use the data used in this manuscript.
- ☐ I certify that the author(s) of the manuscript have documented permission to redistribute/publish the data contained within this replication package. Appropriate permission are documented in the LICENSE.txt file.

Summary of Availability

- ☐ All data **are** publicly available.
- ☐ Some data **cannot be made** publicly available.
- ☒ **No data can be made** publicly available.

INSTRUCTIONS: If *No or some data can be made* **publicly** available, the journal may require that data be preserved for a specified time.

- ☐ Confidential data used in this paper and not provided as part of the public replication package will be preserved for ____ years after publication, in accordance with journal policies.

Dataset list

Data file	Source	Notes	Provided
res_crimes_new2.dta	West Midlands Police	Officer-level incident records: on-scene actions, timestamps, call grade Required variables: arrived_date_time cancelled_date_time crime_number dispatched_date_time gradedresponse incident_date incidentnumber incident_time incident_result int_ref_number left_date_time officer_collar_anon	No
warks1019_new.dta	West Midlands Police	Crime classifications and grid references Required Variables: int_ref_number osgr initial_classification final_classification qualifier	No
caller1017.dta	West Midlands Police	Caller status (victim / third party) Required variables: int_ref_number caller_status	No

Data file	Source	Notes	Provided
<code>callh_vars.dta</code>	West Midlands Police	Call-handler attributes (experience, sex) Required variables: <code>int_ref_number</code> <code>callh_years_service</code> <code>ch_gender</code> <code>call_handlerid</code>	No
<code>crimes_201019_new.dta</code>	West Midlands Police	Crime database with investigation and charge outcomes Required variables: <code>crimenum</code> <code>vict_nominalreturn1</code> <code>vict_cuc_code</code>	No
<code>police_char_dec18.dta</code>	West Midlands Police	Officer characteristics: age, sex, experience Required variables: <code>officer_collar_anon</code> <code>age</code> <code>sex</code> <code>years_service</code>	No
<code>wards_matched_IV.dta</code>	West Midlands Police	Ward assignment of geolocations Required variables: <code>osgr</code> <code>ward_name</code>	No
<code>wards_population.dta</code>	ONS	Ward-level population counts	Yes
<code>ward_housingtype.dta</code>	NOMIS	Dwelling types by ward (fraction flats)	Yes
<code>datefile.dta</code>	Authors	Calendar file	Yes
<code>Analysis_Final_EJ.dta</code>	British Crime Survey	BCS 2005 extract used for Sec. 2.1; requires separate access (see code comment at line 2250)	No

Computational requirements

The code can be run with standard computer hardware and a Stata version 15 or later. The code detects installed software and installs missing packages.

Software Requirements

- ☒ The replication package contains one or more programs to install all

dependencies and set up the necessary directory structure.

- Stata (code was last run with version 18.5)
 - pkgs, ftools, reghdfe, ranktest, ivreghdfe, estout, moremata, mtefe, reghdfe, ivreghdfe, esttab, eststo, mtefe
 - the program “0_install_dependencies.do” is called automatically from the master, checks for existing packages, and will install all dependencies locally if needed.

Controlled Randomness

- ☒ Random seed is set at line 43 in master.do and for each program using random numbers
- ☐ No Pseudo random generator is used in the analysis described here.

Memory, Runtime, Storage Requirements

Summary Approximate time needed to reproduce the analyses on a standard (CURRENT YEAR) desktop machine:

- ☐ <10 minutes
- ☐ 10-60 minutes
- ☐ 1-2 hours
- ☐ 2-8 hours
- ☒ 8-24 hours
- ☐ 1-3 days
- ☐ 3-14 days
- ☐ > 14 days

Approximate storage space needed:

- ☐ < 25 MBytes
- ☐ 25 MB - 250 MB
- ☐ 250 MB - 2 GB
- ☒ 2 GB - 25 GB
- ☐ 25 GB - 250 GB
- ☐ > 250 GB
- ☐ Not feasible to run on a desktop machine, as described below.

Details The code was last run in Stata 18.0 MP Parallel Edition, on Intel(R) Xeon(R) Platinum 8360Y CPU with 2.40GHz, 250 GB RAM, on 64-bit Windows Server 2019 Standard Version 1809.

Description of programs/code

All code is being controlled by `master.do`. Running it prompts `01_build_sample.do` to, which builds the estimation dataset from the raw data files. Then, `02_analysis.do` is started, which produces all figures, tables, and other exhibits in the paper.

- The file `programs/code/01_build_sample.do` will extract and reformat all datasets referenced above. Intermediate data files are save in `Tempdata` folder and the final estimation data set is saved in `Estdata` folder.
- The file `programs/code/02_analysis.do` generates all tables, figures, and other mentions of results in the main body and appendix of the article and are saved in the `Results` folder. Output files are called corresponding names also found in the manuscript.
- The program `programs/code/0_install_dependencies.do` makes sure and installs if necessary all needed dependencies for stata.
- The program `programs/master.do` starts all programs in order.

Instructions to Replicators

- Store the files and folders contained in the replication package in a `yourfolder/replication/` and place the raw data files described above in `yourfolder/Rawdata/`.
- Edit `replication/master.do` to adjust the default path
- Run `replication/master.do` to produce all intermediate data and all results.

Details

- `replication/code/0_install_dependencies.do`:
 - Installs all required Stata packages if they are not already available.
- `replication/code/01_build_sample.do`:
 - Reads the raw data files from the `Rawdata` folder.
 - Cleans, merges, and reformats the data to construct the estimation dataset.
 - Intermediate datasets are stored in the `Tempdata` folder and the final estimation dataset is saved in the `Estdata` folder.
- `replication/code/02_analysis.do`:
 - Produces all tables and figures reported in the main text and appendix.
 - Output is saved in the `Results` folder using filenames corresponding to the manuscript.
 - This program should be run only after `01_build_sample.do` has completed successfully.
- `replication/master.do`:
 - Creates all required output directories.
 - Executes all programs in the correct order: `0_install_dependencies.do`, `01_build_sample.do`, and `02_analysis.do`.

- Running this file reproduces all intermediate datasets, tables, and figures used in the manuscript.

List of tables and programs

The provided code reproduces:

- ☒ All numbers provided in text in the paper
- ☒ All tables and figures in the paper
- ☐ Selected tables and figures in the paper, as explained and justified below.

Figure/Table #	Program	Line Number	Output file
Figure 1	02_analysis.do	130	Figure1_shares_log.txt
Table 1	02_analysis.do	182	Table1.tex
Figure 2	02_analysis.do	232	Figure2.png
Table 2	02_analysis.do	279	Table2a.tex
(Panel A)			
Table 2	02_analysis.do	279	Table2b.tex
(Panel B)			
Table 3	02_analysis.do	494	Table3.tex
Table 4	02_analysis.do	610	Table4.tex
Table 5	02_analysis.do	702	Table5a.tex
(Panel A)			
Table 5	02_analysis.do	702	Table5b.tex
(Panel B)			
Table 6	02_analysis.do	772	Table6.tex
Figure 3	02_analysis.do	860	Figure3.png
Figure A1	02_analysis.do	889	FigureA1.png
Figure A2	02_analysis.do	936	FigureA2.png
Table A1	02_analysis.do	943	TableA1.txt
Table A2	02_analysis.do	975	TableA2.txt
Table A3	02_analysis.do	1167	TableA3a.tex
(Panel A)			
Table A3	02_analysis.do	1167	TableA3b.tex
(Panel B)			
Table A4	02_analysis.do	1248	TableA4.tex
Table A5	02_analysis.do	1322	TableA5.tex
Table A6	02_analysis.do	1434	TableA6.txt
Table A7	02_analysis.do	1459	TableA7.tex
Table A8	02_analysis.do	1719	TableA8.tex
(Sec. 2.3)	02_analysis.do	1883	Text_section23_log.txt
(Footnote 17)	02_analysis.do	1897	Text_footnote17_log.txt
(Sec. 3.1)	02_analysis.do	1897	Text_section31_log.txt
(Sec. 3.2)	02_analysis.do	1897	Text_section32_log.txt

Figure/Table #	Program	Line Number	Output file
(Sec. 4, shares)	02_analysis.do	1933	Text_section4_compliershares_log.txt
(Sec. 4, OLS)	02_analysis.do	1956	Text_section4_complierols_log.txt
(Footnote 24)	02_analysis.do	1998	Text_footnote24.tex
(Sec. 5.2– 5.3)	02_analysis.do	2092	Text_section52_53_reporting.tex
(Footnote 27)	02_analysis.do	2150	Text_footnote27_short.tex
(Sec. 7.1)	02_analysis.do	2184	Text_section71_investshare_log.txt
(Footnote 30)	02_analysis.do	2196	Text_footnote30_log.txt
(Sec. 8, never-taker)	02_analysis.do	2209	Text_section8_nevertaker_log.txt
(Sec. 8, PRTE)	02_analysis.do	860	Text_section8_prte_boot.tex
(Sec. 2.1 BCS numbers)	02_analysis.do	2229	Text_section21_log.txt

References

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Office for National Statistics (ONS). Ward-level Population Estimates. Available at: <https://www.ons.gov.uk>.

NOMIS / Office for National Statistics. Dwelling Types by Ward. Available via the NOMIS web service: <https://www.nomisweb.co.uk>.

Anderberg, Dan, Helmut Rainer, Jonathan Wadsworth, and Tanya Wilson, (2016). Unemployment and domestic violence: Theory and evidence. *The Economic Journal*, 126(597), 1947-1979.